Algorithmic Geometry Fall, 2026

TD 1

1. Determine the discrepancy of the complete set system that contains all subsets of X.

Solution.

For any coloring, there is a set in the complete set system that contains all the elements of the same color — the larger color class has at least $\lceil n/2 \rceil$ elements, so *every* coloring has discrepancy at least $\lceil n/2 \rceil$. Conversely, the balanced coloring ($\lfloor n/2 \rfloor$ elements colored $+1$ and $\lceil n/2 \rceil$ colored -1) satisfies, for every set S,

$$\left|\sum_{v \in S} \chi(v)\right| \;\leq\; \max\{\#(+1\text{-elements}), \#(-1\text{-elements})\} \;=\; \lceil n/2 \rceil.$$

Thus the discrepancy of the complete set system is exactly $\lceil n/2 \rceil$.

2. A graph G can be considered as a special set system in which all sets (edges) have size two. What can you say about the discrepancy of G?

Solution.

If there is no odd-length cycle in G, then G is bipartite. Such a graph has discrepancy 0, since we can color the two parts with the two colors. Otherwise, there is an odd cycle, and so there will be an edge with two endpoints of the same color, giving discrepancy of 2.

3. Let P be a set of n points in $\mathbb{R}$, and $\mathcal{I}$ a set of m intervals.

   Our goal is to find the smallest number of intervals of $\mathcal{I}$ such that each point of P is contained in at least one such interval.

   This is an instance of the set cover problem.

   Show that one can compute the *optimal* solution for this problem.

   Solution.

   We will use a greedy algorithm: sort the points of P from left to right.

   Then, while some point of P is uncovered, let p be the leftmost uncovered point; among the intervals of $\mathcal{I}$ that contain p, take one that covers the maximum number of uncovered points (equivalently, one whose right endpoint is largest). Then repeat this for the uncovered points to get the next interval and so on.

   To prove that our solution is no larger than an optimal solution, do the following:

   Compare the first interval chosen by us to the interval from OPT that covers the left-most point. Our interval chosen must cover all the points of this interval from OPT, since its right endpoint is at least as large and no uncovered point lies left of p.

   Now repeat the above argument starting from the next uncovered point of P.

   Thus our algorithm will get the optimal answer.

4. Consider the following online interval discrepancy game on a set system $(X, \mathcal{R})$, where $X = [n]$ and $\mathcal{R}$ consists of subsets induced by intervals on $[n]$:

   At iteration $t \in [n]$, the adversary gives a new point in $[n]$.

   Your goal is to color it immediately with a -1 or a $+1$ color.

   Give a strategy for the adversary such that no matter how you color the points of X, there will be an interval with discrepancy $\Omega(\sqrt{n})$ at some point during the game.

Solution. Set $c \in (1/\sqrt{n}, 1/4]$ to be a fixed constant, say $c = 1/10$ (valid for all sufficiently large n; assume for simplicity that all quantities below are integers).

Let S_j be the (signed) sum of all the colored elements after j iterations. If at any iteration this sum becomes greater than $c\sqrt{n}$ or less than $-c\sqrt{n}$, then clearly the interval $[n]$ gives the required lower bound. So assume that $S_j \in [-c\sqrt{n}, +c\sqrt{n})$ for all $j \in [n]$.

As the value of the sum changes by $+1$ or -1 at each iteration, either at least $n/2$ iterations are up-moves ($S_{j+1} = S_j + 1$) or at least $n/2$ are down-moves; suppose the former (the other case is symmetric). By the pigeonhole principle over the $2c\sqrt{n}$ possible values of the sum, there exists a value s such that $S_j = s$ and $S_{j+1} = s + 1$ for $r \geq \frac{n/2}{2c\sqrt{n}} = \sqrt{n}/4c$ iterations $j \in [n]$.

In particular, the signed sum of all the elements of $[n]$ colored *during those iterations where the total sum is at most s*, must be at least

$$r - 2c\sqrt{n} \geq \sqrt{n}\left(\frac{1}{4c} - 2c\right) = \Omega(\sqrt{n}) :$$

the r elements above are colored $+1$, and each element colored -1 while the sum is at most s must be paid for by an earlier element colored $+1$ at a strictly smaller sum level, except for at most one -1 per level, i.e. at most $2c\sqrt{n}$ in total.

We'd be done if we could ensure that for any s, all the elements of $[n]$ colored while the sum is at most s, are in some interval. That is easy: map each value of the sum s to a set of $\sqrt{n}/2c$ consecutive points in $[n]$, sequentially over the values of s. That is, at each iteration j, if $S_j = t$, the adversary gives a new point from the interval

$$\Big[\left(t + c\sqrt{n}\right)\sqrt{n}/2c, \left(t + c\sqrt{n}\right)\sqrt{n}/2c + \sqrt{n}/2c\Big).$$

These intervals partition $[n]$ (a total of $2c\sqrt{n} \cdot \sqrt{n}/2c = n$ points). If at any iteration a block becomes full, then the sum has visited the corresponding value t at least $\sqrt{n}/2c$ times; at least half of these visits are followed by an up-move, or at least half by a down-move, and the same argument (or its mirror image, using the blocks with index $\geq t$) gives an interval with $\Omega(\sqrt{n})$ discrepancy.

5. For the previous question, give a strategy that achieves discrepancy $O(\sqrt{n})$ at all times.

Solution.

Partition $[n]$ into $\sqrt{n}$ consecutive intervals of length $\sqrt{n}$. For each such canonical interval, as the new points arrive that belong to it, we will alternately color them with -1 and $+1$. So at all times, the signed sum of the points in the same canonical interval is -1, 0 or $+1$.

For any other interval I, if I fully contains a canonical interval, that adds $\{-1, 0, +1\}$ to its discrepancy, for a total of $\sqrt{n}$ value. On the other hand, the (at most) two canonical intervals containing the two endpoints of I can add a further $2\sqrt{n}$ to its discrepancy. This gives a total discrepancy of $O(\sqrt{n})$.

6. For the experts problem, assume that there exists an expert who is *always* correct. Then give a simple deterministic algorithm that outputs a prediction for each event t, such that

$$\text{total mistakes by our algorithm} \leq \log n.$$

Solution. We follow the majority prediction (by count). At the end of each iteration, we remove all the experts that predicted incorrectly for that iteration, and continue on to the next iteration with the remaining experts.

As one expert is never wrong, this expert will never be removed. Further, each time we predict incorrectly, the majority of experts must have been wrong, and so we reduce the number of experts by half. This can only happen at most $\log n$ times.

7. Prove the following algorithm makes at most $(\mathrm{OPT}+1)(\lfloor \log_2 n \rfloor + 1)$ mistakes, where OPT is the number of mistakes made by the best expert in $[n]$.

   We follow the majority prediction (by count). At each iteration, we remove all the experts that were wrong for that iteration, and continue on to the next iteration with the remaining experts.

   If we ever run out of experts, we re-start with all the n experts.

   Solution. Consider one *phase*: the run of the algorithm between two consecutive restarts, starting with all n experts alive. Each of our mistakes discards the majority side, i.e. at least half of the alive experts, so after ℓ mistakes at most $n/2^{\ell}$ experts remain alive. A phase can therefore end (the alive set becomes empty) only after $\ell > \log_2 n$ mistakes, i.e. a completed phase costs at most $\lfloor \log_2 n \rfloor + 1$ mistakes.

   If a phase ends, the best expert must have made a mistake during it — otherwise it would never have been discarded. Hence the number of completed phases is at most OPT.

   The final phase never empties, so at least one expert is always alive; each mistake still halves the alive count, which stays ≥ 1, so the final phase costs at most $\lfloor \log_2 n \rfloor$ mistakes.

   Thus in total, we made at most

$$\mathrm{OPT}\left(\lfloor \log_2 n \rfloor + 1\right) + \lfloor \log_2 n \rfloor \;\leq\; (\mathrm{OPT}+1)\left(\lfloor \log_2 n \rfloor + 1\right) \;=\; O\big((\mathrm{OPT}+1)\log n\big)$$

   mistakes.

8. Over each of the m events, construct the predictions of n experts, and an adversary which can choose the outcome of each event, to prove that any deterministic algorithm must make at least $\lfloor \log_2 n \rfloor$ more errors than the best expert. (Assume $m \geq \lfloor \log_2 n \rfloor$.)

Solution. (Assume n is a power of two for simplicity; otherwise split the experts as evenly as possible.) The adversary maintains a set of *clean* experts — those with no mistake so far — initially all n experts.

In the first iteration, half the clean experts predict 1, and the other half predict 0. After we have made a prediction, the adversary will choose the outcome that makes our prediction wrong. This fixes the best expert to lie in one of the two halves that predicted correctly.

We now continue with the $n/2$ clean experts that predicted correctly in the first iteration, and repeat the same argument.

After k rounds at least $n/2^k$ clean experts remain and we have made k mistakes. As long as at least two clean experts remain, i.e. for all $k \leq \lfloor \log_2 n \rfloor - 1$, the next round forces one more mistake. Thus after $\lfloor \log_2 n \rfloor$ rounds, we have made $\lfloor \log_2 n \rfloor$ mistakes, while the best expert has made no mistake.

9. Over each of the m events, construct the predictions of n experts, and an adversary which can choose the outcome of each event, to prove that any deterministic algorithm must make at least twice as many errors as the best expert.

Solution. There are two experts, one predicting 1 and the other predicting 0, at each iteration.

The adversary will choose the outcome such that our prediction is always wrong.

Then the best expert makes at most $m/2$ mistakes, while we make m mistakes.

10. Consider the following more general setup.

    Let $[n]$ be a set of n experts. At each iteration $t = 1, \ldots, m$, each expert makes a prediction for the t-th iteration. After the expert predictions, the loss of each expert is revealed. That is, there is a loss function

    $$\ell^t : \{1, \ldots, n\} \to [0, 1]$$

    that assigns a *loss* to the prediction of every expert in the t-th iteration.

    Then show that for any deterministic algorithm, an adversary can choose the losses ℓ^t such that the total loss of the algorithm is at least n times the loss of the best expert.

Solution. The adversary sets the loss of the expert $i^t \in [n]$ that the algorithm follows at iteration t as 1, and the losses of all other experts are set to 0.

Then clearly the algorithm has total loss m. On the other hand, by the pigeonhole principle, there is an expert that was chosen at most m/n times by the algorithm; so its total loss is at most m/n.